

DB에 넣을 것

admins

- 관리자 포털 로그인용 계정
- 비밀번호는 **해시저장**

email_accounts

- 이메일 계정 정보
- 이메일 주소, 표시명, 부서, 활성화 여부
- 비밀번호가 필요하다면 **평문 말고 해시저장**

audit_logs

- 누가 로그인했고 어떤 계정을 만들었는지 기록

WSL2에 MariaDB 설치

설치

sudo apt update

sudo apt install -y mariadb-server

```
apogatto@DESKTOP-LEHJIUV:~/SME_ATTACK_LAB$ sudo apt update
[sudo] password for apogatto:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1875 kB]
Ign:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1663 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1875 kB]
Fetched 3662 kB in 37s (99.1 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
96 packages can be upgraded. Run 'apt list --upgradable' to see them.
apogatto@DESKTOP-LEHJIUV:~/SME_ATTACK_LAB$ sudo apt install -y mariadb-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  galera-4 libcgi-fast-perl libcgi-pm-perl libclone-perl libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl
  libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl
  libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmariadb3
  libmysqlclient21 libncurses6 libnuma1 libsnappy1v5 libterm-readkey-perl libtimedate-perl liburi-perl liburing2
  libwrap0 mariadb-client mariadb-client-core mariadb-common mariadb-plugin-provider-bzip2 mariadb-plugin-provider-lz4
  mariadb-plugin-provider-lzma mariadb-plugin-provider-lzo mariadb-plugin-provider-snappy mariadb-server-core
  mysql-common pv socat
Suggested packages:
  libmldbm-perl libnet-daemon-perl libsql-statement-perl libdata-dump-perl libipc-sharedcache-perl
```

실행

WSL2에서 systemd가 켜져 있으면:

sudo systemctl enable mariadb

sudo systemctl start mariadb

```
apogatto@DESKTOP-LEHJIUV:~/SME_ATTACK_LAB$ sudo systemctl enable mariadb
Synchronizing state of mariadb.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mariadb
apogatto@DESKTOP-LEHJIUV:~/SME_ATTACK_LAB$ sudo systemctl start mariadb
```

결과값 없는게 정상.

db 초기 보안 설정

sudo mysql_secure_installation

```
apogatto@DESKTOP-LEHJIUV:~/SME_ATTACK_LAB$ sudo mysql_secure_installation
```

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):

OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] y

Enabled successfully!

Reloading privilege tables..

... Success!

현재 root 비밀번호: 엔터

unix_socket 인증 사용: Y 또는 N 둘 다 가능

root 비밀번호 변경: Y

익명 사용자 제거: Y

원격 root 로그인 차단: Y

test DB 제거: N

권한 테이블 재적용: Y

DB 시작 명령어

sudo mariadb -u root -p

```
apogatto@DESKTOP-LEHJIUV:~/SME_ATTACK_LAB$ sudo mariadb -u root -p
```

Enter password:

Welcome to the MariaDB monitor. Commands end with ; or \g.

Your MariaDB connection id is 41

Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

DB SQL (그대로 복붙)

```
CREATE DATABASE admin_portal CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
```

```
CREATE USER 'portal_user'@'%' IDENTIFIED BY 'StrongPortalPass!123';
GRANT ALL PRIVILEGES ON admin_portal.*TO 'portal_user'@'%';
FLUSH PRIVILEGES;
```

```
USE admin_portal;
```

```
CREATE TABLE admins (
  id INT AUTO_INCREMENT PRIMARY KEY,
  username VARCHAR(50) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  full_name VARCHAR(100) NOT NULL,
  role ENUM('superadmin', 'admin', 'viewer') NOT NULL DEFAULT 'admin',
  is_active TINYINT(1) NOT NULL DEFAULT 1,
  last_login_at DATETIME NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

```
CREATE TABLE email_accounts (
  id INT AUTO_INCREMENT PRIMARY KEY,
  email_address VARCHAR(255) NOT NULL UNIQUE,
  login_id VARCHAR(100) NOT NULL UNIQUE,
  display_name VARCHAR(100) NOT NULL,
  department VARCHAR(100) NULL,
  password_hash VARCHAR(255) NOT NULL,
  mailbox_quota_mb INT NOT NULL DEFAULT 1024,
  is_active TINYINT(1) NOT NULL DEFAULT 1,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

```
CREATE TABLE audit_logs (
  id BIGINT AUTO_INCREMENT PRIMARY KEY,
  admin_id INT NULL,
  action VARCHAR(100) NOT NULL,
  target_type VARCHAR(50) NULL,
  target_id INT NULL,
  ip_address VARCHAR(45) NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_audit_admin
    FOREIGN KEY (admin_id) REFERENCES admins(id)
    ON DELETE SET NULL
);
```

이후 php로 해시 생성하기 위해 php를 설치

sudo apt install php -y

```
apogatto@DESKTOP-LEHJ1UV:~/SME_ATTACK_LAB$ sudo apt install php -y
[sudo] password for apogatto:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64 liblua5.4-0 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache
  php8.3-readline ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser ufw php-pear
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64 liblua5.4-0 php php-common php8.3 php8.3-cli php8.3-common php8.3-opcache
```

설치 후 기존 관리자 비밀번호를 입력 시 해시 값 생성

```
apogatto@DESKTOP-LEHJ1UV:~/SME_ATTACK_LAB$ php -r "echo password_hash('qhdksdchlrlh', PASSWORD_DEFAULT) . PHP_EOL;"
$2y$10$LBHy6xblHwNmB/5ruJaRV.hR4R15Aym7TIR..HCUt591Db2yf87Je
```

이후 DB 다시 접속 후 계정 등록

```
INSERT INTO admins (username, password_hash, full_name, role, is_active)
VALUES(
  'admin',
  '**여기에_위에서_생성한_해시값**',
  'System Admin',
  'superadmin',
  1
);
```

삽입 내용 확인

select * from admins;

```
MariaDB [admin_portal]> select * from admins;
+----+-----+-----+-----+-----+-----+-----+-----+
| id | username | password_hash | full_name | role | is_active | last_login_at | created_at | updated_at |
+----+-----+-----+-----+-----+-----+-----+-----+
| 1 | admin | $2y$10$LBHy6xblHwNmB/5ruJaRV.hR4R15Aym7TIR..HCUt591Db2yf87Je | System Admin | superadmin | 1 | NULL | 2026-04-04 18:24:38 | 2026-04-04 18:30:27 |
+----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)
```

php와 DB를 연결하기 위한 서비스 설치

sudo apt install php-mysql -y

sudo service apache2 restart

여기서 잠깐

우린 nginx를 80포트로 쓰고있기에 apache2와 포트가 겹침
그래서 apache2를 8080으로 옮기는 것을 추천

sudo nano /etc/apache2/ports.conf

Listen 80을 8080으로 수정

```
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 8080
```

사이트 설정도 같이 수정

sudo nano /etc/apache2/sites-available/000-default.conf

```
<VirtualHost *:8080>
# The ServerName directive sets the request scheme, hostname and port that
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
#ServerName www.example.com

ServerAdmin webmaster@localhost
DocumentRoot /var/www/html
```

맨 위 virtualhost를 8080으로

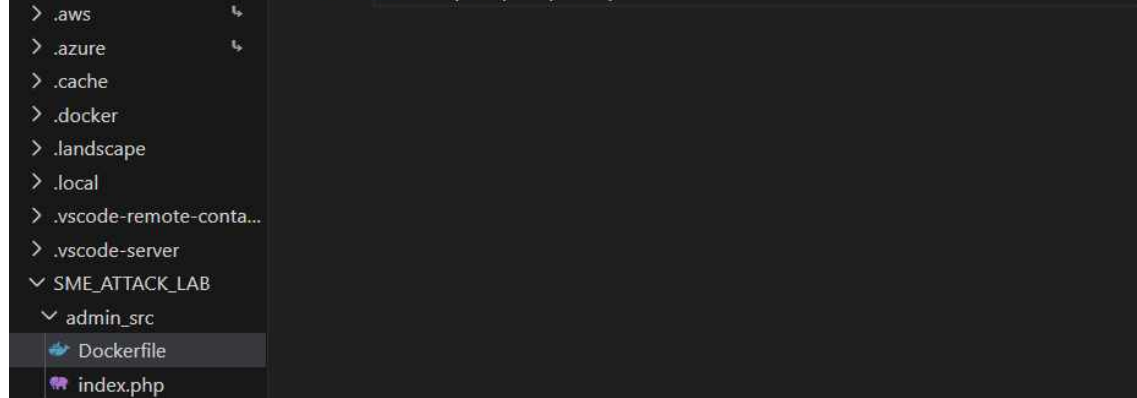
파일 수정은 제가 작성한거 올리긴 할텐데 어디가 바뀌었는
지 표시만 하겠습니다.

수정되는 파일

- Dockerfile (추가됨)
- docker-compose.yml (이미지 사용에서 내부 파일 빌드로 수정)

이제 admin_antry를 다시 빌드하기 위한 Dockerfile을 추가

```
FROM php:7.4-apache
RUN docker-php-ext-install pdo pdo_mysql mysqli
COPY . /var/www/html/
```



docker-compose.yml 수정

```
services:
  # --- 1. 관리자 포털 (PHP로 변경) ---
  admin_entry:
    build: ./admin_src
    container_name: admin_entry
    ports:
      - "8003:80"
    volumes:
      - ./admin_src:/var/www/html
    extra_hosts:
      - "host.docker.internal:host-gateway"
    environment:
      DB_HOST: host.docker.internal
      DB_NAME: admin_portal
      DB_USER: portal_user
      DB_PASS: StrongPortalPass!123
    networks:
      sme_internal_network:
        ipv4_address: 192.168.20.40
```

기존 admin and try 컨테이너 삭제 후 재빌드
docker compose up -d --build admin_entry

빌드 후 해당 검색이 나오면 성공

docker exec -it admin_entry php -m | grep -E 'pdo_mysql|mysqli'

```
apogatto@DESKTOP-LEHJIUV:~/SME_ATTACK_LAB$ docker exec -it admin_entry php -m | grep -E 'pdo_mysql|mysqli'
mysqli
pdo_mysql
```


그럼 이제 포털가서 로그인 해보고 정상적으로 로그인 되는지 확인

Secure-Project 통합 관리자 포털

Vault 연동 · 컨테이너 관리 데모

로그인: admin [로그아웃](#)

운영 중인 컨테이너

목록에서 컨테이너를 선택해 성능을 조회하거나 Vault에서 키/내부 정보를 가져올 수 있습니다.

Client A - Juice Shop
id: client_a - 192.168.20.10:3000
CPU: 26% · MEM: 38% · 응답: 62ms
[검속\(내부정보\)](#) [지표 갱신](#)

Client B - DVWA
id: client_b - 192.168.20.20:80
CPU: 57% · MEM: 63% · 응답: 38ms
[검속\(내부정보\)](#) [지표 갱신](#)

Client C - bWAPP
id: client_c - 192.168.20.30:80
CPU: 56% · MEM: 44% · 응답: 180ms
[검속\(내부정보\)](#) [지표 갱신](#)

최근 액션 / 내부 정보

7:07:25 PM - client_c 지표 갱신
7:07:25 PM - client_b 지표 갱신
7:07:25 PM - client_a 지표 갱신
아직 작업이 없습니다.

Vault 설정

Vault 주소: <http://192.168.20.50:8200/v1/secret/data/>
(토론은 서버 환경변수로 관리됩니다)

전체 제어

[지표 전체 갱신](#) [모두 중지\(시뮬레이션\)](#)

로그 확인

```
sudo mariadb -u root -p
select id, username, last_login_at from admins;
```

```
+-----+-----+-----+
| id | username | last_login_at |
+-----+-----+-----+
| 1 | admin | 2026-04-04 19:02:10 |
+-----+-----+-----+
1 row in set (0.000 sec)
```

```
SELECT * FROM audit_logs ORDER BY id DESC;
```

```
MariaDB [admin_portal]> SELECT * FROM audit_logs ORDER BY id DESC;
+-----+-----+-----+-----+-----+-----+
| id | admin_id | action | target_type | target_id | ip_address | created_at |
+-----+-----+-----+-----+-----+-----+
| 1 | 1 | LOGIN_SUCCESS | admin | NULL | 192.168.20.1 | 2026-04-04 19:02:10 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)
```